

Monte Carlo Methods

Introduction

Generating Samples

Markov Chain Monte Carlo

Improving Efficiency

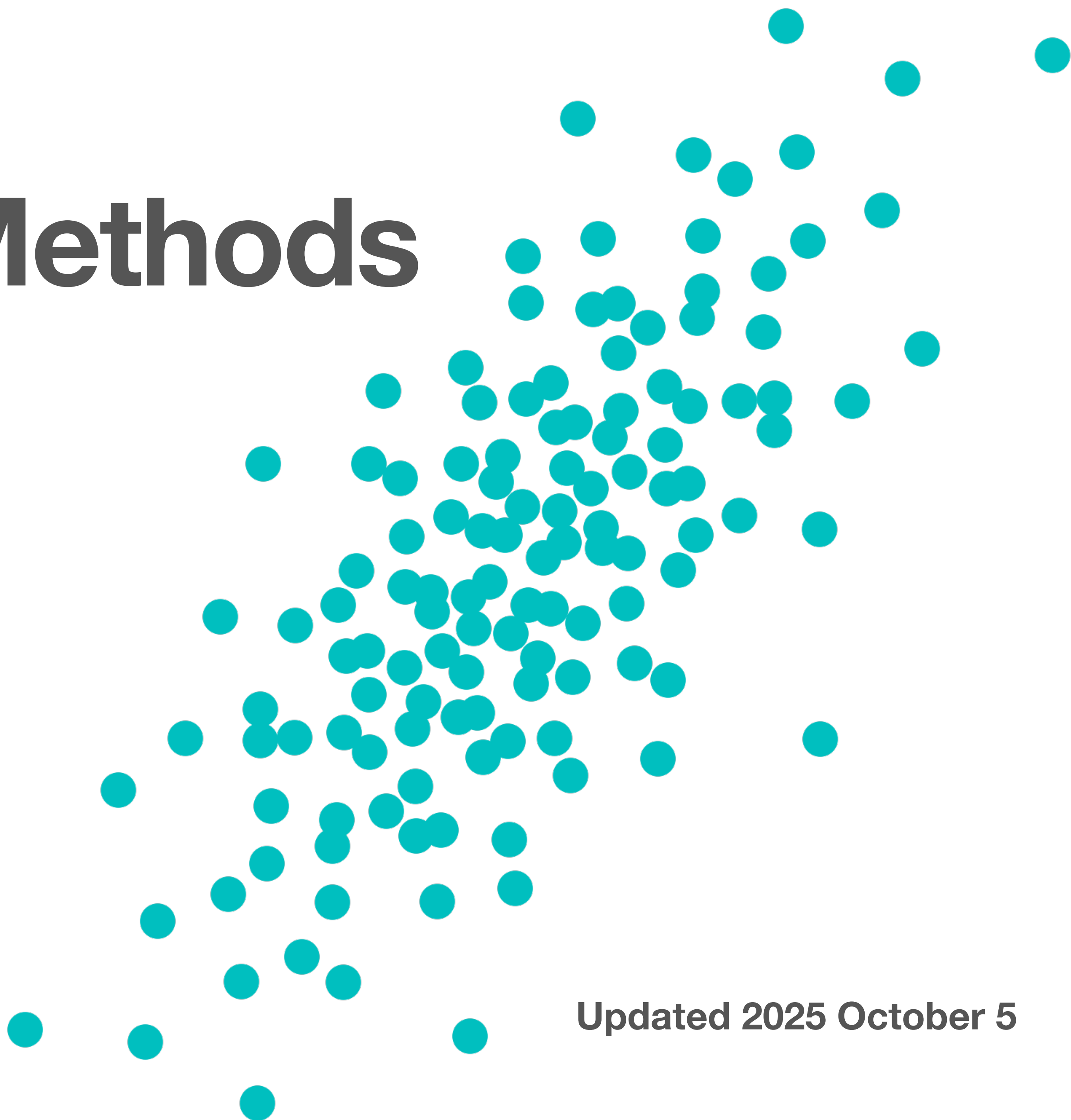
Selected Topics

Git website and repository

Canvas

Fred Hickernell, Fall 2025

Updated 2025 October 5

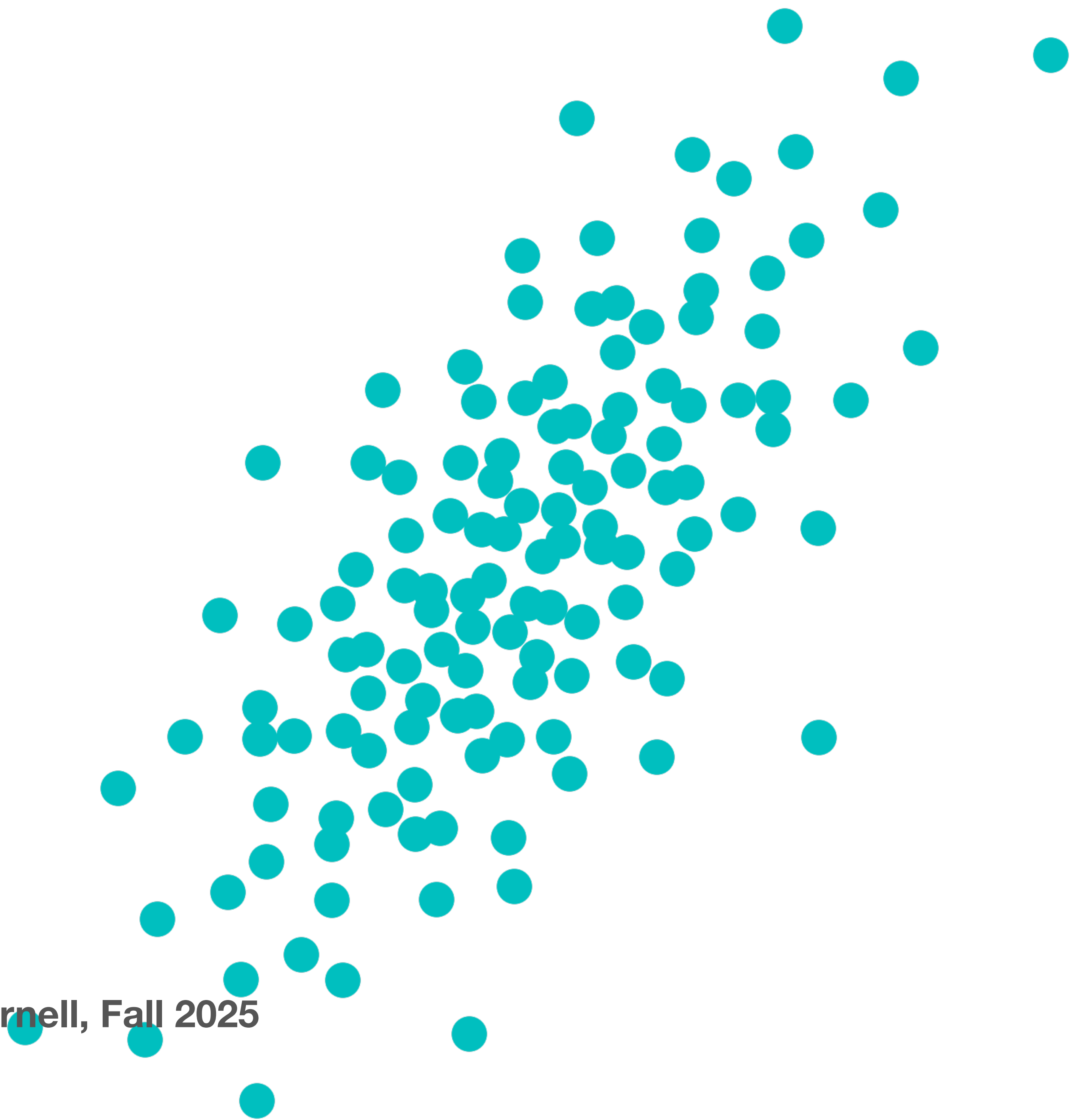


Introduction

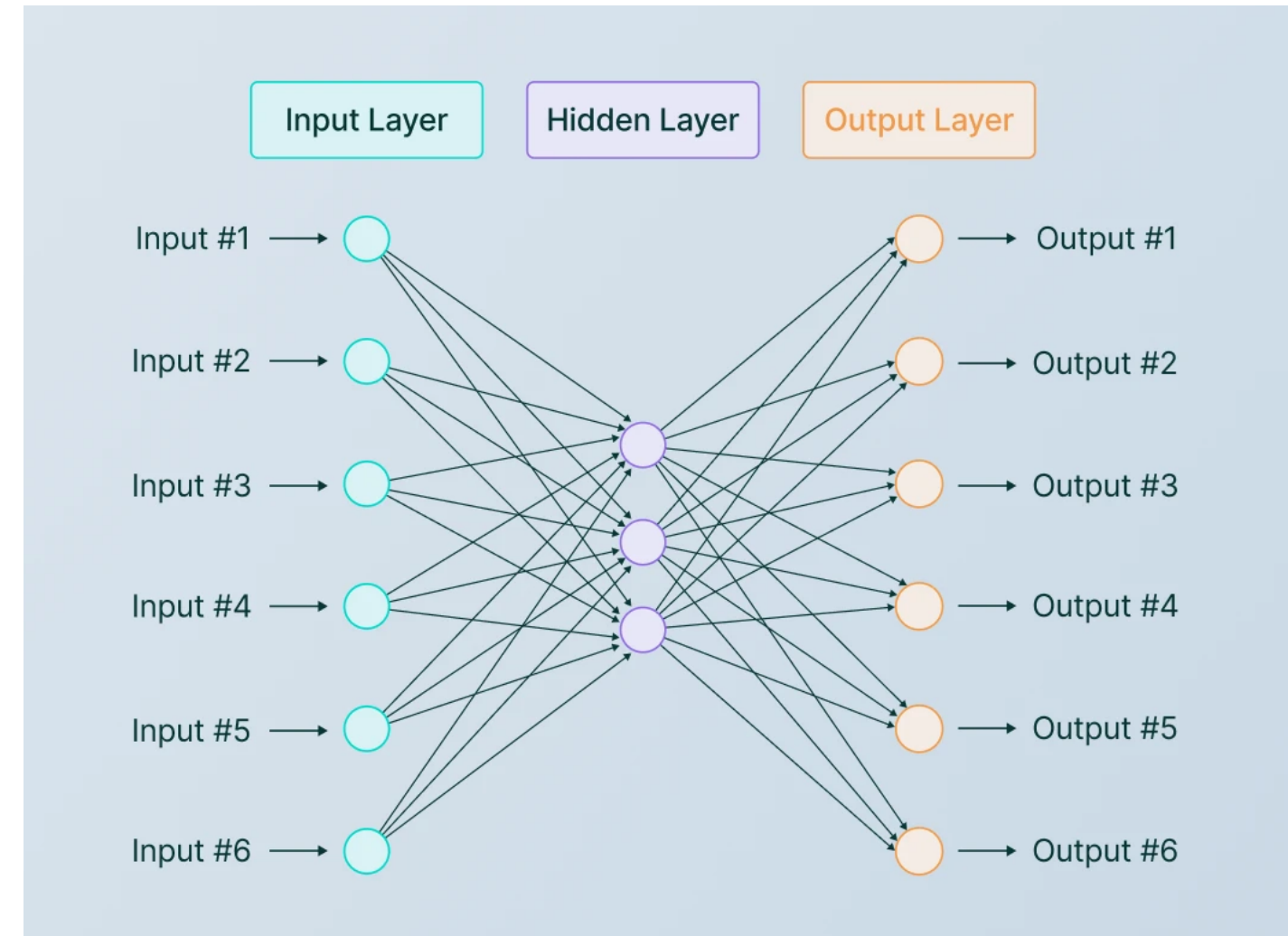
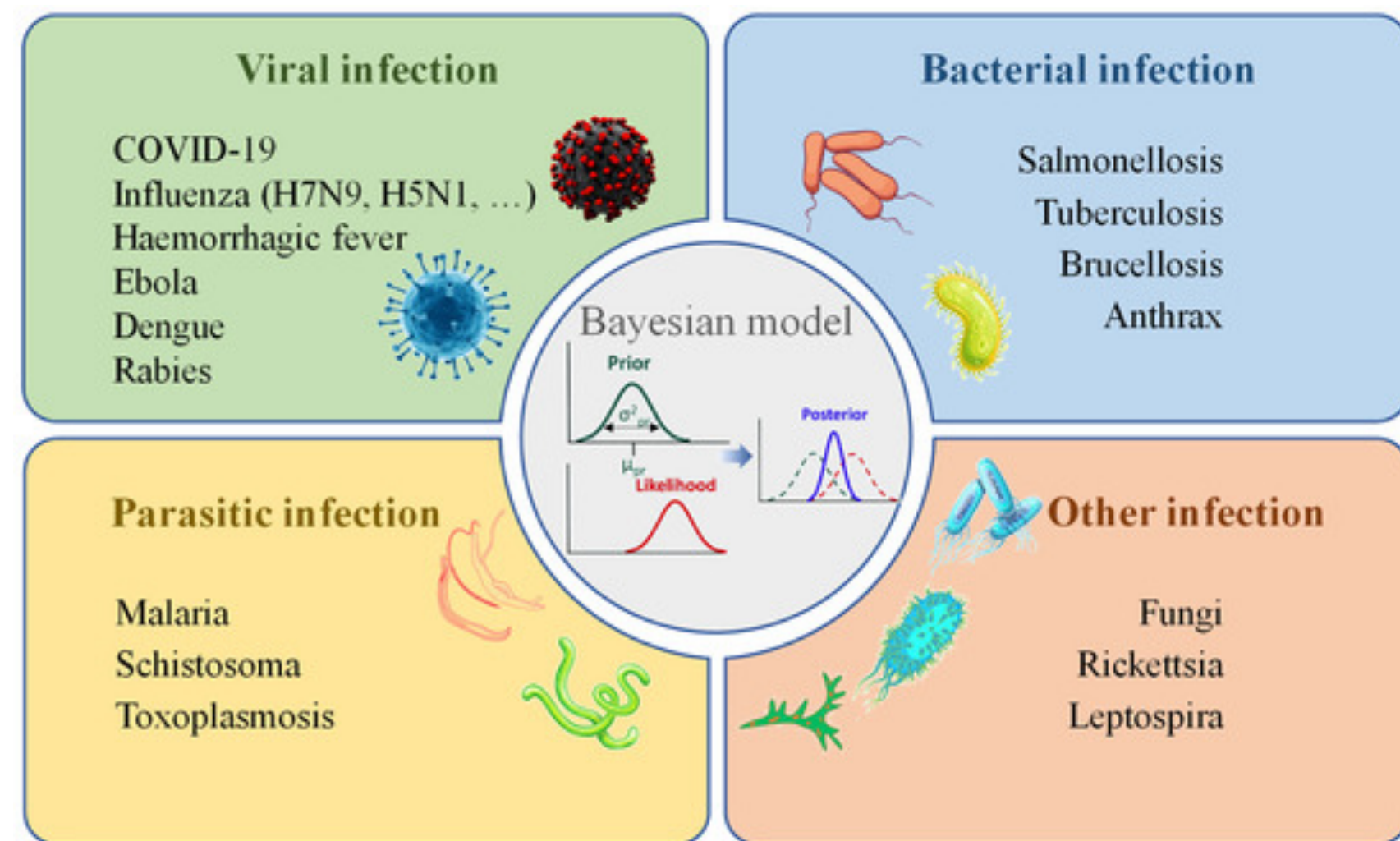
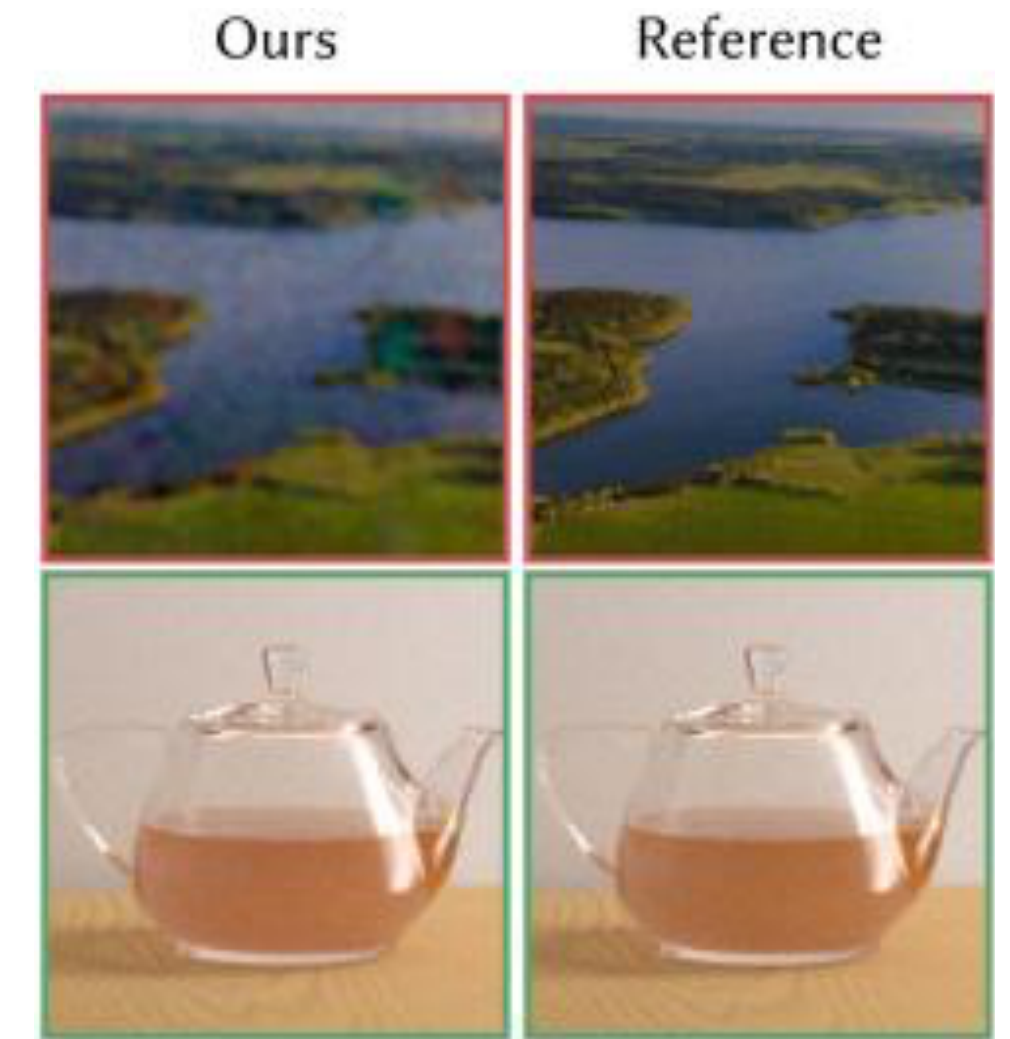
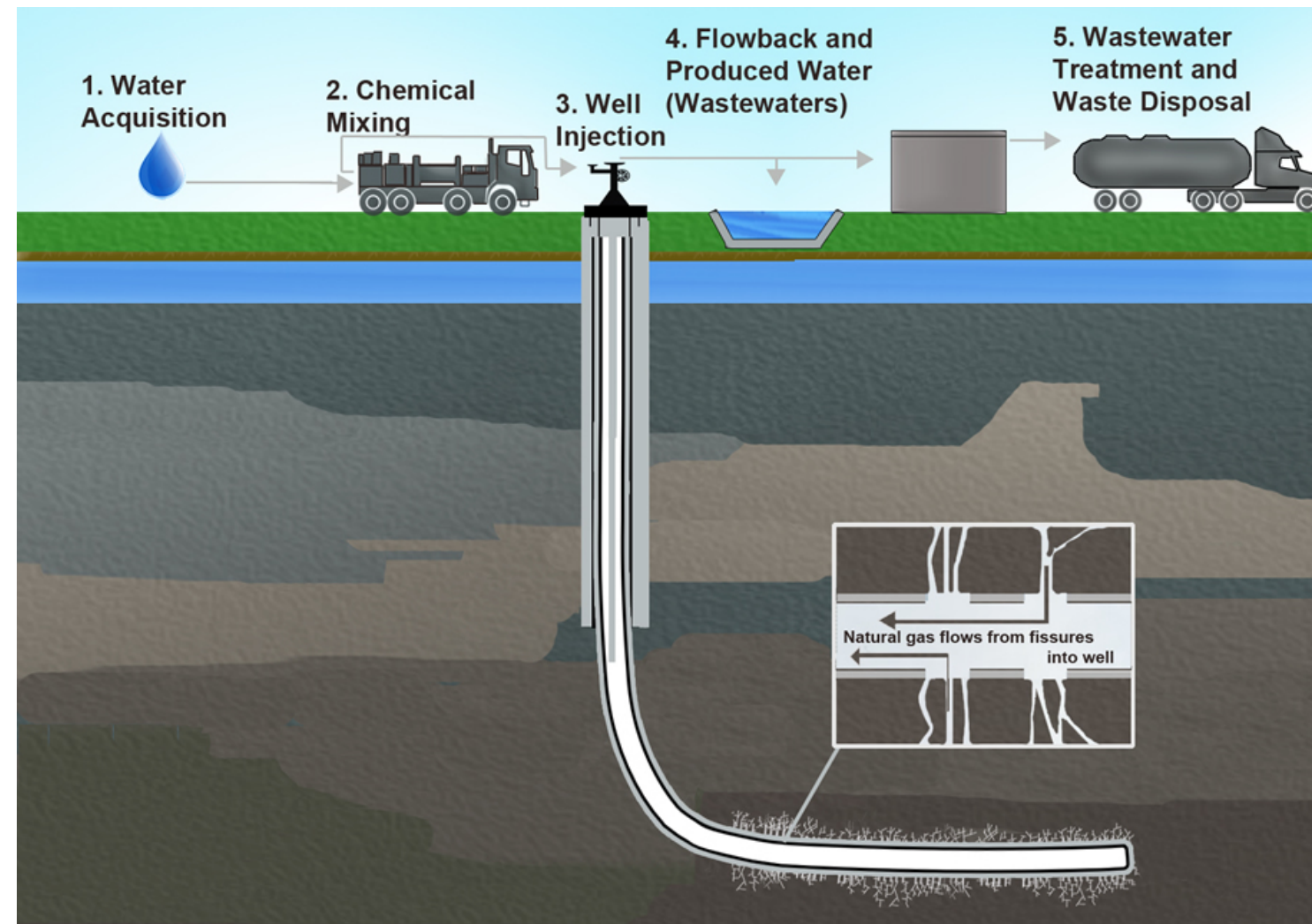
Owen, Chapters 1 & 2

Assignment 1 Due Sep 5

MATH 565 Monte Carlo Methods, Fred Hickernell, Fall 2025



Monte Carlo Helps With Uncertainty [WSJ 2016]





Why is there uncertainty?



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- Finance — **market forces**, often modeled by stochastic processes driven by Brownian motion



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- Queues — **arrival times and service times** of customers



How is this expressed quantitatively?

Y = random variable denoting quantity of interest = {
option payoff
fluid pressure
pixel intensity
statistical model parameter
neural network parameter
service time

= $f(\mathbf{X})$, where

$\mathbf{X}$ = multivariate random variable with a simpler distribution

We want to estimate the mean, variance, quantile, or probability distribution of Y
using sample versions



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How long should you plan for the trip to take?

Let's look at this Jupyter Notebook [AreWeThereYet](#) on the [class website](#)



Let me know your degree program?

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Why should you attend synchronously?

- You will be better keep **pace**?
- You will get **real time** answers to questions?
- You can **influence** the pace and direction of the course?
- You can help your peers learn and benefit from them—partake in a **learning community**
- To help me know you, in case you want a **reference**, or to add me to your **LinkedIn** network



Review of probability

Let Y be a **random variable** with a sample space $\mathcal{Y} \subseteq \mathbb{R}$

$\mathbb{P}(Y \in \Omega)$ means the probability of the **event** Ω

F is the **cumulative distribution function** $F(y) := \mathbb{P}(Y \leq y), y \in \mathcal{Y}$

Q is the **quantile function** $Q(p) := \inf\{y \in \mathcal{Y} : F(y) \geq p\}, 0 < p < 1$

$\mathbb{E}[g(Y)] := \int_{\mathcal{Y}} g(y) dF(y)$ is the **expectation** of $g(Y)$

discrete

continuous

ϱ is the **probability mass function**

ϱ is the **probability density function**

$$\varrho(y) := \mathbb{P}(Y = y)$$

$$\varrho(y) := F'(y)$$

$$\mathbb{E}[g(Y)] = \sum_{y \in \mathcal{Y}} g(y) \varrho(y)$$

$$\mathbb{E}[g(Y)] = \int_{y \in \mathcal{Y}} g(y) \varrho(y) dy$$



Moments

$\mu := \mathbb{E}(Y)$ is the **mean** of Y

$\sigma^2 = \text{var}(Y) := \mathbb{E}[(Y - \mu)^2]$ is the **variance** of Y

σ is the **standard deviation**

$$\text{cov}(\mathbf{Y}) := \left(\mathbb{E}[(Y_j - \mu_j)(Y_k - \mu_k)] \right)_{j,k=1}^d$$

is the **covariance matrix** of the random **vector** $\mathbf{Y}$



Binomial distribution

This is a coin-flipping distribution with the probability p of heads

$$Y \sim \text{Binomial}(n, p) \quad \text{scipy.stats.binom}(n, p)$$

$$\varrho(y) := \mathbb{P}(Y = y) = \binom{n}{y} p^y (1 - p)^{n-y}, \quad y = 0, 1, \dots, n$$

$$F(y) = \mathbb{P}(Y \leq y) = \sum_{j=0}^{\lfloor y \rfloor} \binom{n}{j} p^j (1 - p)^{n-j}$$

$$Q(p) = \inf\{y \in \{0, 1, \dots, n\} : F(y) \geq p\}$$

$$\mu = \mathbb{E}(Y) = np \quad \sigma^2 = \text{var}(Y) = np(1 - p)$$

The **Bernoulli** distribution is the case $n = 1$ and $\mu = p$



Exponential distribution

This distribution models the waiting time for the next customer or taxi, assuming that the time to the next one is independent of how long you have waited so far

$$Y \sim \text{Exponential}(\lambda) \quad \text{scipy.stats.expon(loc=0, scale=1/\lambda)}$$

$$\varrho(y) := \begin{cases} 0, & -\infty < y < 0 \\ \lambda \exp(-\lambda y), & 0 \leq y < \infty \end{cases}$$

$$F(y) = \mathbb{P}(Y \leq y) = \begin{cases} 0, & -\infty < y < 0 \\ 1 - \exp(-\lambda y), & 0 \leq y < \infty \end{cases}$$

$$Q(p) = \frac{-\log(1-p)}{\lambda}, \quad 0 < p < 1$$

$$\mu = \mathbb{E}(Y) = 1/\lambda \quad \sigma^2 = \text{var}(Y) = 1/\lambda^2$$



Multivariate normal/Gaussian distribution

Used in finance and uncertainty quantification

$Y \sim \mathcal{N}(\mu, \Sigma)$ is a d -dimensional random vector

`scipy.stats.multivariate_normal(mean, cov)`

μ = mean

Σ = covariance matrix

$$\varrho(\mathbf{y}) := \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{y} - \mu)^\top \Sigma^{-1}(\mathbf{y} - \mu)\right)$$

$$F(\mathbf{y}) = \int_{(-\infty, \mathbf{y})} \varrho(\mathbf{z}) \, d\mathbf{z}$$



Sampling

Let Y be a random variable, often $Y = f(\mathbf{X})$

$Y_1, Y_2, \dots$ be a **sample**

not necessarily random or independent and identically distributed (IID)

$\hat{\mu}_n := \frac{1}{n} \sum_{i=1}^n Y_i$ is the **sample mean**

$\hat{\sigma}_n^2 := \frac{1}{n-1} \sum_{i=1}^n (Y_i - \hat{\mu}_n)^2$ is the **sample variance**



Under **random** sampling

Let Y be a random variable, often $Y = f(\mathbf{X})$

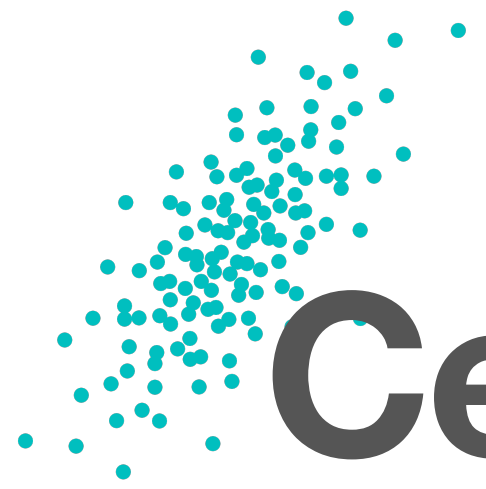
$$Y_1, Y_2, \dots \sim Y$$

$$\mathbb{E}(\hat{\mu}_n) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}(Y_i) = \mu \text{ so } \hat{\mu}_n \text{ is unbiased}$$

If in addition, $Y_1, Y_2, \dots$ are **uncorrelated**, then

$$\mathbb{E}(\hat{\sigma}_n^2) = \sigma^2 \text{ so } \hat{\sigma}_n^2 \text{ is unbiased}$$

$$\text{mse}(\hat{\mu}_n) := \mathbb{E}[(\mu - \hat{\mu}_n)^2] = \text{var}(\hat{\mu}_n) = \frac{\sigma^2}{n}, \quad \text{rmse}(\hat{\mu}_n) = \frac{\sigma}{\sqrt{n}}$$



Central Limit Theorem

Let Y be a random variable, often $Y = f(\mathbf{X})$

$$Y_1, Y_2, \dots \stackrel{\text{iid}}{\sim} Y$$

The distribution of

$$\frac{\hat{\mu}_n - \mu}{\sigma / \sqrt{n}} \text{ approaches } \mathcal{N}(0, 1) \text{ as } n \rightarrow \infty$$

which means that

$$\hat{\mu}_n \dot{\sim} \mathcal{N}(\mu, \sigma^2/n) \text{ for large } n$$



Monte Carlo estimates of properties of Y

Let Y be a random variable and let $Y_1, Y_2, \dots$ be an IID random sample

Population Quantity	Sample Approximation
mean μ	$\hat{\mu}_n \pm 2.58\hat{\sigma}_{n_0}^2$, <code>np.mean(data)</code>
variance σ^2	$\hat{\sigma}_n^2$, <code>np.var(data, ddof=1)</code>
cumulative distribution function F	empirical distribution function $\hat{F}_n$ <code>statsmodels.distributions.empirical_distribution.ECDF(data)</code>
quantile function Q	empirical quantile function $\hat{Q}_n$ <code>np.quantile(data, probabilities)</code>
probability density or mass function ϱ	histogram, <code>np.histogram(data, bins = "auto")</code> kernel density estimator, <code>scipy.stats.gaussian_kde(data)</code>

For μ we have an **interval** estimator, but for the others typically **point** estimators



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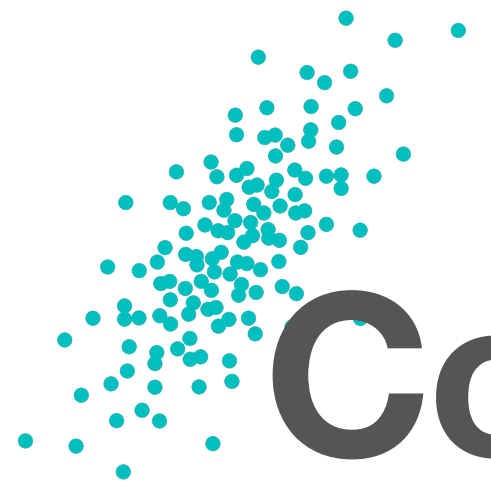
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Conditional probability

Let (Y, Z) be a vector of random variables

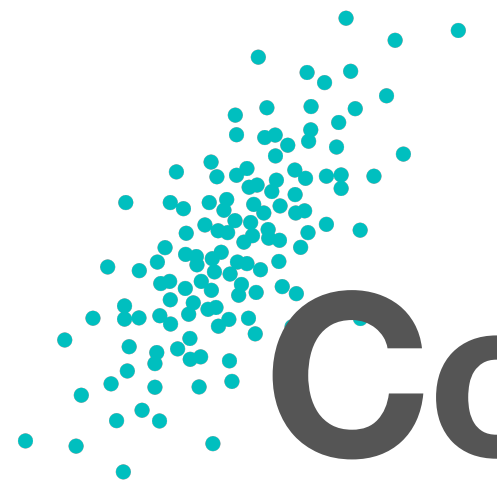
$$\mathbb{P}(Y = y \mid Z = z) = \frac{\mathbb{P}(Y = y \& Z = z)}{\mathbb{P}(Z = z)}$$

$$\mathbb{P}(Y = y \mid Z = z) = \frac{\mathbb{P}(Z = z \mid Y = y)\mathbb{P}(Y = y)}{\mathbb{P}(Z = z)} \quad \text{Bayes' Theorem}$$

$$\varrho_{Y|Z}(y \mid z) = \frac{\varrho_{Y,Z}(y, z)}{\varrho_Z(z)}$$

$$\mathbb{E}(Y) = \mathbb{E}_Z[\mathbb{E}_Y(Y \mid Z)]$$

$$\text{var}(Y) = \mathbb{E}_Z[\text{var}_Y(Y \mid Z)] + \text{var}_Z[\mathbb{E}_Y(Y \mid Z)]$$



Conditional Monte Carlo

Let $(Y = f(\mathbf{X}_{1:d}), \mathbf{X}_{1:d})$ be a vector of random variables

Let $\mathbf{Z} = \mathbf{X}_{2:d}$

If $g(\mathbf{X}_{2:d}) = \mathbb{E}_Y(Y \mid \mathbf{X}_{2:d})$ has an analytic form, then

$$\mu = \mathbb{E}(Y) = \mathbb{E}_{\mathbf{X}_{2:d}}[g(\mathbf{X}_{2:d})] \text{ and } \text{var}(Y) \geq \text{var}_{\mathbf{Z}}[g(\mathbf{X}_{2:d})]$$

If $h(y, \mathbf{X}_{2:d}) = \varrho_{Y \mid \mathbf{X}_{2:d}}(y \mid \mathbf{x}_{2:d})$ has an analytic form, then

$$\varrho_Y(y) = \mathbb{E}_{\mathbf{X}_{2:d}}[h(y, \mathbf{X}_{2:d})] \quad \text{probability density expressed as expectation}$$



Big Ideas

- **Sample quantities**, often using IID samples, used to estimate population properties of a random variable with a complex distribution, including
 - Means, variances, covariances
 - Probability density/mass functions, cumulative distribution functions, and quantile functions
- For means we have **interval** estimators, which indicate the uncertainty in our estimates, using the CLT
- For the other properties we have **point** estimators, i.e., only one approximate value for each thing to be estimated